

# Nuclear gluonometry with a tensor-polarized ${}^6\text{Li}$ beam at the Electron-Ion Collider: projected inclusive and coherent $\cos 2\phi$ measurements

C. Peng

Physics Division, Argonne National Laboratory

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**Abstract.** The double-helicity-flip structure function  $\Delta(x, Q^2)$  of a spin-1 target is generated at leading twist by gluons alone and has never been measured. We project its measurement at the Electron-Ion Collider with a transversely tensor-polarized  ${}^6\text{Li}$  beam, in inclusive deep inelastic scattering, where only the scattered electron is detected, and in coherent diffraction with the  ${}^6\text{Li}$  recoiling intact. With the leading-order bag-model moment of Sather and Schmidt ported to  ${}^6\text{Li}$  as a labelled scenario, the inclusive  $\cos 2\phi$  amplitudes at the four most sensitive  $(x, Q^2)$  bins,  $x = 0.011\text{--}0.14$  and  $Q^2 = 1.1\text{--}14 \text{ GeV}^2$ , are  $(0.44\text{--}0.95)\times 10^{-2}$  at  $P_{zz} = 0.6$ , against single-state truth-level statistical errors of  $(1.4\text{--}4.5)\times 10^{-4}$  in one EIC year ( $10 \text{ fb}^{-1}$  per nucleon): a  $21\text{--}44\sigma$  measurement per bin whose  $x$  dependence discriminates between moment-constrained interpretations, and an extraction of  $x\Delta(x, Q^2)$  itself to  $2\text{--}9\%$  in the best bins. At the reconstructed level, with the spin-state-sorted ratio of  $m = \pm 1$ -rich and  $m = 0$ -rich bunches and a hadronic-method  $y$  from a PYTHIA 8 final state, the errors are  $(0.9\text{--}3.0)\times 10^{-4}$  and the fit closes on the response-folded truth. The coherent channel carries the same modulation with two separable mechanisms — a nuclear-shape term anchored on the polarized-deuteron colour-glass calculation, predicted to flip sign for  ${}^6\text{Li}$ , and a flat exotic-gluon term — and  ${}^6\text{Li}$ 's anomalously small quadrupole moment makes it a null test. Its intact recoil keeps the beam rigidity and, with the Yellow Report's beam divergences, cannot be tagged at IP6 with any published optics; the channel defines a far-forward requirement — a dedicated tagging optics or the second interaction region's secondary focus — and for a reference tag with a  $0.20 \text{ GeV}$  near-beam envelope at the high-acceptance luminosity — which no IP6 optics provides — one year would yield  $1.7\times 10^7$  events and a  $5\sigma$  sensitivity to an amplitude of  $0.9\%$  per unit  $P_{zz}$ , against a shape term of  $6\%$  and an exotic-gluon term of  $0.3\text{--}1\%$  on the same footing. A tagging optics at IP6 — the horizontal  $\beta^*$  raised  $50\text{--}180\times$  with the pots following the envelope, the vertical plane untouched — recovers a  $32\text{--}42\%$  tag at  $1/7\text{--}1/13$  of the luminosity:  $2.6\text{--}6.1\times 10^6$  tagged events per year, an  $8\text{--}11\sigma$  measurement of the shape term per year and  $5\sigma$  on a  $1\%$  exotic-gluon term in  $3\text{--}4$  years,  $3\text{--}8\times$  below what a second interaction region's secondary focus would deliver at the same luminosity.

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## 1 · Introduction

A spin-1 target can absorb two units of photon helicity in the forward Compton amplitude. The structure function that measures this,  $\Delta(x, Q^2)$ , receives no leading-twist contribution from quarks or from bound spin- $\frac{1}{2}$  nucleons, so in a nucleus it is direct evidence for glue that is not inherited from the individual nucleons — "exotic glue" in the language of Jaffe and Manohar's nuclear gluonometry [1,2]. It has never been measured for any target; the one prior experimental document is a JLab letter of intent for a  ${}^{14}\text{N}$  target, without a sensitivity projection [3]. The only absolute normalizations available are moment estimates: a bag-model calculation for the spin- $3/2$   $\Delta^{++}$  baryon [4], maximal-transversity scenarios for proton–deuteron Drell–Yan [5], and lattice-QCD matrix elements for the  $\phi$  meson [6] and the deuteron [7]. The EPIOS white paper on polarized ion beams at the EIC names spin-1  ${}^6\text{Li}$  among the nuclei where a discovery is most likely [8], and the companion physics case [9] argues why  ${}^6\text{Li}$  — a polarized deuteron bound to an inert  $\alpha$  particle, with  $A/Z = 2$  and deuteron-like spin dynamics in the ring — is the practical choice.

This paper presents the first numerical projection of a  $\Delta$  measurement for any target. Two channels are considered. In inclusive DIS on a transversely tensor-polarized  ${}^6\text{Li}$  beam the scattered electron's azimuth about the alignment axis acquires a  $\cos 2\phi$  modulation proportional to  $\Delta$ . In coherent diffraction,  $e {}^6\text{Li} \rightarrow e' X {}^6\text{Li}(\text{g.s.})$ , the nucleus stays in its ground state and the modulation of the tagged yield mixes the gluonic term with a nuclear-shape term that the polarized-deuteron calculation of Mäntysaari et al. [10] predicts — and  ${}^6\text{Li}$ , with a quadrupole moment fifty times smaller than  ${}^7\text{Li}$ 's, separates the two. Section 2 states the observable, Section 3 the simulation framework, Section 4 the  $\Delta$  model, Section 5

the inclusive projections at truth and reconstructed level, Section 6 the coherent channel with the far-forward requirement it imposes, Section 7 the systematics and assumptions.

## 2 • The observable

For unpolarized electrons on a spin-1 target whose spin axis makes angle  $\theta_m$  with the beam, the inclusive cross section per spin projection  $m$  is [1,2]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[ F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left( F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_m \Delta(x, Q^2) \cos 2\phi \right]$$

$$a_m = \frac{1}{4} c_m (3 \cos^2 \theta_m - 1), \quad c_m = 3|\lambda_m| - 2 \rightarrow (1, -2, 1)$$

with  $\phi$  the azimuth of the lepton plane about the virtual photon, measured from the transverse projection of the spin axis. Averaged over a fill,  $c_m$  becomes the tensor polarization  $P_{zz} = p_+ + p_- - 2p_0$ . We write the  $b_1, b_2$  terms with the sign of [2] as transcribed in the companion physics case [9]; in the Hoodbhoy–Jaffe–Manohar convention as applied by HERMES and by Cosyn et al. [11] they carry the opposite sign, which affects only the sign of  $A_{zz}$  relative to  $b_1$  and the constant term of the spin-state ratio of §3.3, nothing in the  $\Delta$  sector. At leading order  $\Delta$  is generated by the gluon transversity distribution  $\delta G$  through the  $O(\alpha_s)$  photon–gluon convolution

$$\Delta(x, Q^2) = -\frac{\alpha_s(Q^2)}{2\pi} \left( \sum_q e_q^2 \right) x^2 \int_x^1 \frac{dy}{y^3} \delta G(y, Q^2)$$

in the sign convention of Ref. [6]. For a transverse spin axis ( $\theta_m = 90^\circ$ ) the measured amplitude and its inversion to the structure function are

$$A^{\cos 2\phi} = -\frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2, \quad \hat{\Delta} = -\hat{A} \frac{y^2 D_\phi}{1-y}$$

Three properties of this relation shape the analysis. First, with  $F_2 = 2x(1+R)F_1$  the inversion becomes  $\Delta/F_1 = -\hat{A}[2(1+R) + y^2/(1-y)]$ , so at the  $y \lesssim 0.03$  of the sensitive region  $F_2$  cancels and  $R = \sigma_L/\sigma_T$  is the only structure-function input that reaches  $\Delta/F_1$ . Second, the amplitude is exactly linear in  $\Delta$  (§5.3). Third, the target-mass corrections are small: at the four bins of §3.4,  $y^2 = 4M^2 x^2/Q^2 \leq 0.006$ , so the approximation set of the simulation —  $A_{\parallel} = D \cdot g_1/F_1$  with the  $y^2$ -suppressed term dropped,  $g_2 = g_2^{WW}$ , no target-mass terms — is adequate. One finite- $y$  effect is physical rather than a labelling matter: the tensor-polarized  $b_1$ – $b_3$  terms feed the  $\cos 2\phi$  amplitude at  $O(y^2)$ , bounded by  $1.15 y^2 b_1/F_1$  (the full Cosyn et al. Eq. 17e combination,  $\approx 6.9\times$  the  $y^2 b_1/6$  of the  $b_1$  term alone) [11]; for a HERMES-like  $b_1$  this is  $\leq 1\%$  of the projected amplitudes at the four bins and is computable from the same data's  $A_{zz}$ .

The size of  $\Delta$  is unknown to within an order of magnitude. The bag-model moment of Sather and Schmidt,  $\int x \Delta dx = -0.012 \alpha_s$  with its  $Q^2$  evolution but no  $x$  dependence [4], puts the asymmetry at the per-mille level; the explicit  $\alpha_s(Q^2)$  is the leading-order Wilson coefficient of the photon–gluon box, so the  $O(\alpha_s)$  suppression relative to  $F_1$  is model-independent while  $-0.012$  is the bag estimate. Lattice QCD adds two results, both at unphysical pion masses: the bare  $\phi$ -meson matrix element  $A_2 = 0.23(2)(5)$  at  $m_\pi = 450$  MeV [6], which converts to  $\int x \Delta|_\phi \approx -0.006 \alpha_s$  (our conversion, unrenormalized), and the NPLQCD deuteron  $\delta G$  moment of  $-0.010(3)$ , bare at  $m_\pi \approx 806$  MeV [7] — a transversity-to-momentum ratio roughly ten times below the  $\phi$ 's, implying the opposite sign for the deuteron's  $\Delta$  moment. Percent-level asymmetries arise only in the maximal-transversity ( $\Delta_T g = \Delta g$ ) Drell–Yan scenarios their authors call optimistic [5]. Nothing exists for  ${}^6\text{Li}$  or any  $A > 2$  nucleus beyond a  $\Delta\Delta$ -isobar estimate for the deuteron [12], so the literature brackets the nuclear  $\Delta$  in both directions: suppressed, as in the NPLQCD deuteron, or enhanced by binding in analogy with the EMC effect [8]. The projections therefore carry the moment-constrained model of §4 together with the scenario band  $\Delta/F_1 \in 10^{-3}$ – $10^{-2}$ .

## 3 · Simulation framework

### 3.1 · Beams, polarization and luminosity

EIC ions are  $\gamma$ -matched to the proton rather than rigidity-scaled, because the two rings must share a revolution period, so the  ${}^6\text{Li}$  menu is 40.8 GeV/u and 99.5–137.5 GeV/u, the top value being the rigidity cap  $275 \text{ GeV} \times Z/A$  [8,13]. The projections are made at the mid configuration,  $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV/u})$ ,  $\sqrt{s_{\text{NN}}} = 63 \text{ GeV}$ , with the low ( $5 \times 40.8$ ) and top ( $18 \times 137.5$ ) configurations used where the hadronic reconstruction or the far-forward acceptance depends on the energy. The in-ring tensor polarization is a placeholder  $P_{\text{zz}} = 0.6$  (the ANL source targets  $P_{\text{zz}} \geq 0.8$  [9]; transport to the ring is an open accelerator question, and every statistical error scales as  $1/P_{\text{zz}}$ ), the alignment axis is the vertical — the stable spin direction of the arcs, native at the interaction point — and the electrons are unpolarized. Luminosity is quoted per nucleon:  $10 \text{ fb}^{-1}/\text{u}$ , "one EIC year" in the Yellow Report's accounting (30 weeks at  $10^{33} \text{ cm}^{-2} \text{ s}^{-1}$  with 60% efficiency [13]), and  $100 \text{ fb}^{-1}/\text{u}$  for the ten-year programme, all of it in the transverse-tensor fill. No lithium luminosity has been published; the yields scale linearly with whatever the machine delivers.

### 3.2 · Generator and structure functions

No public generator produces polarized  $e+A$  events for  $A > 1$ : BeAGLE is unpolarized, DJANGO and PEPSI are nucleon-level and longitudinal-only. The generator used here, polligen, implements the missing component — the spin-density matrix  $\otimes$  structure-function master formula with spin-correlated sampling — and takes the hadronic final state from a PYTHIA 8 production of 8 M events over the three beam configurations. It is validated against the analytic asymmetry formulas and by pseudo-experiments, including unbiased  $\cos 2\phi$  recovery under  $\phi$ -acceptance holes; a full-luminosity pseudo-experiment draws each  $\phi$ -bin count from a Poisson distribution with the exact expected yield, which is statistically identical to event-level sampling.

Structure functions are toy parameterizations with PDF-grid backends (CT18NLO, EPPS21, NNPDFpol11) behind one interface, so rates hold to a factor  $\approx 1.5$  (the toy  $F_2$  is within  $\pm 37\%$  of CT18 over the acceptance) and asymmetry shapes are scenarios by construction.  $R = \sigma_L/\sigma_T$  is a placeholder of the right magnitude, 0.14–0.18 at the four bins of §3.4, in every projection below; the published SLAC/E143 R1998 world fit [14], all three of its functional forms, sits behind the same hook, and switching to it moves  $\Delta/F_1$  by +18, +19, +8 and –4% at the four bins, with a further 0.5–6% from the spread of the fit forms — a correction to apply rather than a band, and the systematic on every  $\Delta$  quoted below until it is.

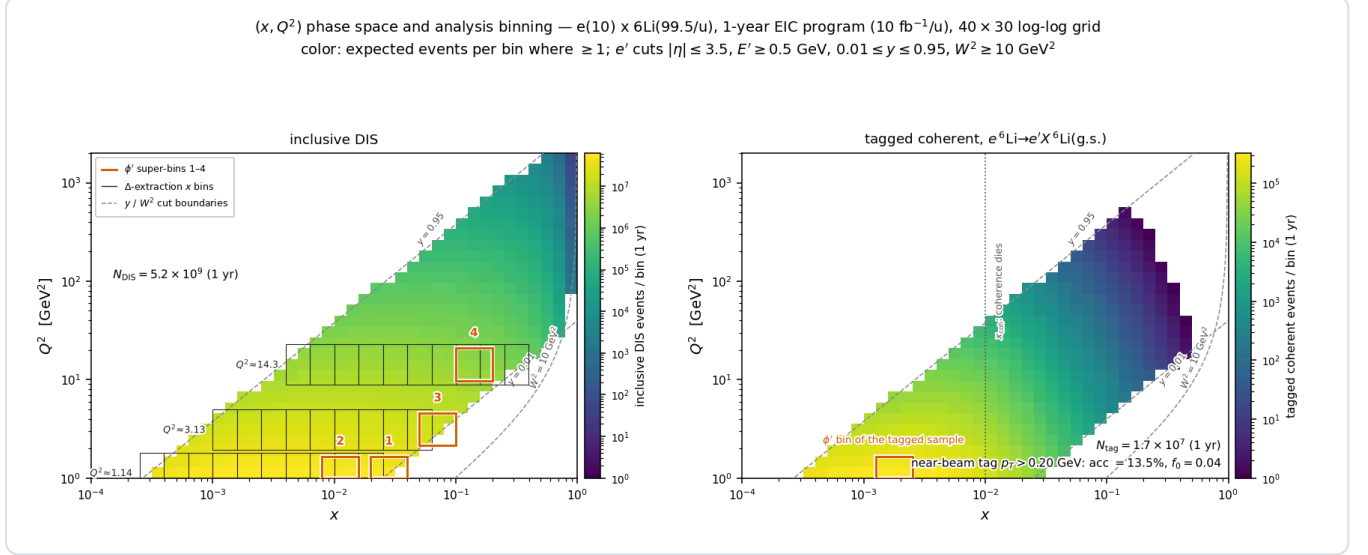
### 3.3 · Estimators

The figures use a single spin state: a binned least-squares fit of  $N(\phi') = C(1 + A \cos 2\phi')$  rather than the  $2\langle \cos 2\phi' \rangle$  moment, which realistic  $\phi$  holes bias by more than the signal while the fit stays unbiased, with  $\delta A = \sqrt{(2/N)/P_{\text{zz}}}$  validated by pseudo-experiments and no  $\cos \phi'$  term, since the  $m$ -symmetric tensor fills set it to zero by construction. The azimuth  $\phi'$  is the covariant one built from the four-vectors [15], equal to  $\phi_e - \phi_S$  for a massless target and within 5 mrad of it for  ${}^6\text{Li}$ , and the 25 mrad crossing angle changes only odd harmonics of the electron azimuth because the ePIC axis is the electron beam.

The measurement itself cannot use a single state: the fit then also measures the detector's own  $\cos 2\phi'$  acceptance harmonic divided by  $P_{\text{zz}}$ , and a 3% harmonic aligned with the vertical spin axis fakes an amplitude of 0.05, five to eleven times the signal. The estimator is therefore the spin-state-sorted ratio of  $m = \pm 1$ -rich ( $P_{\text{zz}} = +0.6$ ) and  $m = 0$ -rich ( $-1.2$ ) bunches. It cancels any acceptance common to both states bin by bin, turns relative-luminosity errors into a constant, and gives, with  $N$  the total of both states,  $\delta \hat{A} = 2\sqrt{(2/N)/(P_+ - P_0)} = 0.67 \times$  the single-state error if the  $m = 0$ -rich fill reaches  $|P_{\text{zz}}| = 1.2$ . The cancellation of the acceptance, however, holds only for an acceptance common to both states: for two fills the bias is  $(\varepsilon_2^+ - \varepsilon_2^0)/(P_+ - P_0)$  at any luminosity split, so a  $10^{-3}$  difference of the  $\cos 2\phi'$  harmonic fakes  $5.6 \times 10^{-4}$ , 6–13% of the projected amplitudes and 2–6 one-year statistical errors. Bunch-by-bunch alternation, or  $10^{-4}$  stability of the harmonic between fills, is thus a requirement of the measurement rather than a refinement; with it, a relative-luminosity error between the states enters only as a constant.

### 3.4 • Acceptance and binning

Rates use per-nucleon luminosity,  $F_2^A/A$ , and scattered-electron cuts  $|\eta| \leq 3.5$ ,  $E' \geq 0.5$  GeV,  $0.01 \leq y \leq 0.95$  and  $W^2 \geq 10$  GeV<sup>2</sup>, which accept  $5.2 \times 10^9$  events in one year (Figure 1). The four displayed  $(x, Q^2)$  super-bins are selected by statistical significance within  $Q^2$  bands — an unconstrained ranking concentrates at the lowest  $Q^2$ , whereas the  $Q^2$  lever arm provides the consistency check against power-suppressed contributions — and each merges  $3 \times 3$  cells of the  $40 \times 30$  analysis grid ( $3 \times 2$  for the two  $Q^2 = 1.14$  GeV<sup>2</sup> bins, clipped at the grid's  $Q^2 = 1$  GeV<sup>2</sup> edge). They sit at  $(x, Q^2) = (0.028, 1.14)$ ,  $(0.011, 1.14)$ ,  $(0.071, 3.13)$  and  $(0.141, 14.3$  GeV<sup>2</sup>), i.e. at  $y = 0.010, 0.026, 0.011$  and  $0.026$ : the amplitude grows as  $(1 - y)/y^2$ , so the sensitivity lives at the lowest  $y$  the detector can reconstruct, which is what makes the kinematic reconstruction of §5.2 the central detector question.



**Figure 1** — phase-space coverage and analysis binning at  $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV/u})$ , one EIC year,  $40 \times 30$  log-log grid; colour: expected events per bin where  $\geq 1$ ; dashed:  $y$  and  $W^2$  cut boundaries. **Left, inclusive DIS:** boxes 1–4 are the  $\phi'$  super-bins of Figure 2; grey combs are the three  $Q^2$  slices ( $Q^2 \times [1/1.6, 1.6]$ ) whose merged  $x$ -bin pairs give the data points of the amplitude-versus- $x$  panel and of the  $\Delta$  extraction (Figures 2 and 3), retained where  $N \geq 10^3$  and  $\delta A \leq 8 \times 10^{-3}$ . **Right, tagged coherent channel:** the same map times  $f_{\text{coh}}(x)$  and the tag acceptance of §6.2, almost entirely at  $x < 0.03$ ; the box is the super-bin of Figure 5(d).

## 4 • The $\Delta$ model

The default model, A, is an ansatz normalized to the bag moment applied to the per-nucleon  $\Delta$  of  ${}^6\text{Li}$ :

$$\Delta_A(x, Q^2) = \frac{1}{3} A \alpha_s(Q^2) F_1(x, Q^2) \frac{x^a(1-x)^b}{N_{\text{peak}}}, \quad \int_0^1 x \Delta(x, \langle Q^2 \rangle) dx = -0.012 \alpha_s \Rightarrow A = -0.292$$

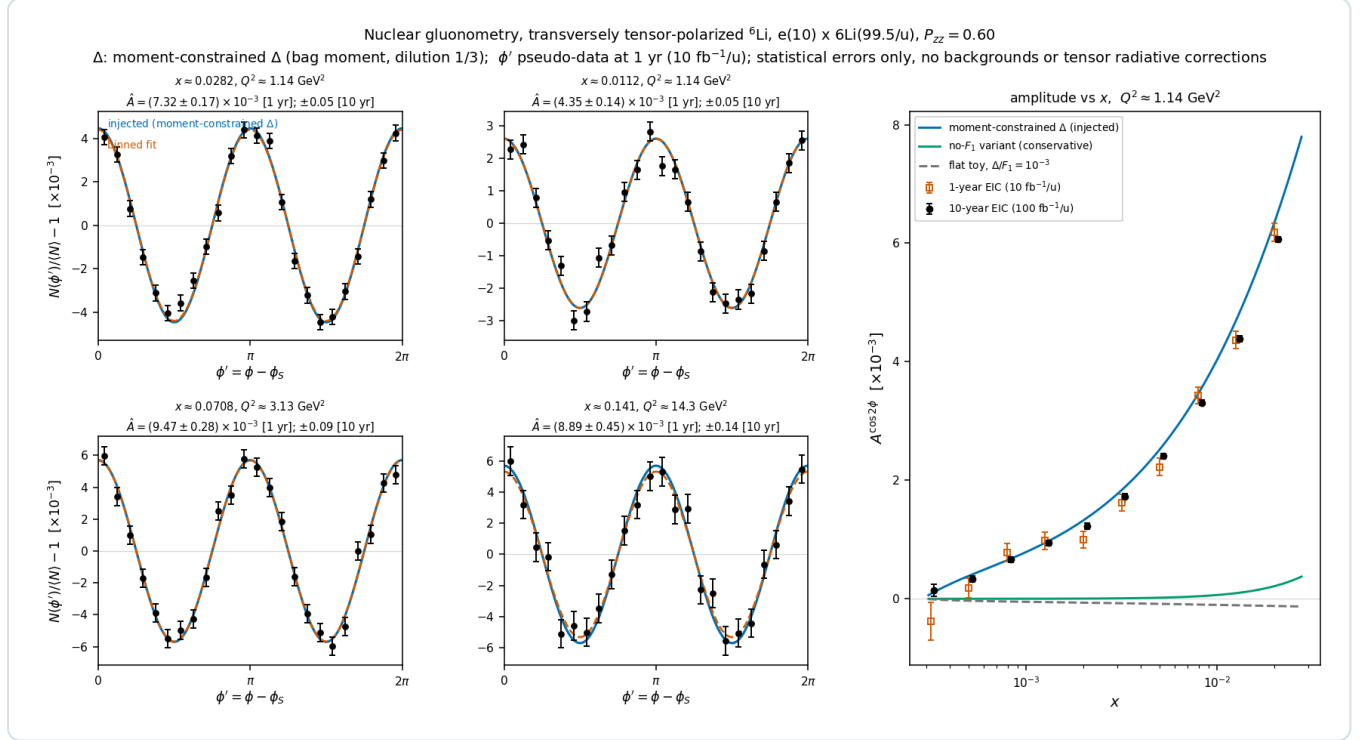
with  $(a, b) = (0.7, 3)$ . The factor  $1/3$  restricts the exotic glue to the polarized deuteron-like pair while the whole-nucleus  $P_{zz}$  remains a separate factor, so nothing is double counted; the cluster picture would give  $\approx 0.81$  instead, an upside of  $\times 2.4$  that is not taken. The constraint is imposed at the rate-weighted  $\langle Q^2 \rangle = 4.47$  GeV<sup>2</sup> of the accepted phase space, and because  $F_1$  evolves the realized moment varies by a few per cent across the plotted  $Q^2$  slices;  $\alpha_s$  comes from the CT18NLO table.

Model B imposes the same constraint without the  $F_1$  factor and yields amplitudes a factor 3–56 lower at the four bins (20–56 in the  $Q^2 = 1.14$  GeV<sup>2</sup> bins, 3–8 at the higher  $Q^2$ ); the A-versus-B spread is the dominant ansatz systematic, and both curves appear on every plot. Porting a single-hadron bag moment to a per-nucleon nuclear structure function has no precedent, so normalization and  $x$  dependence are both scenario choices, bracketed by model B from below and possible nuclear enhancement from above.

## 5 • Inclusive projections

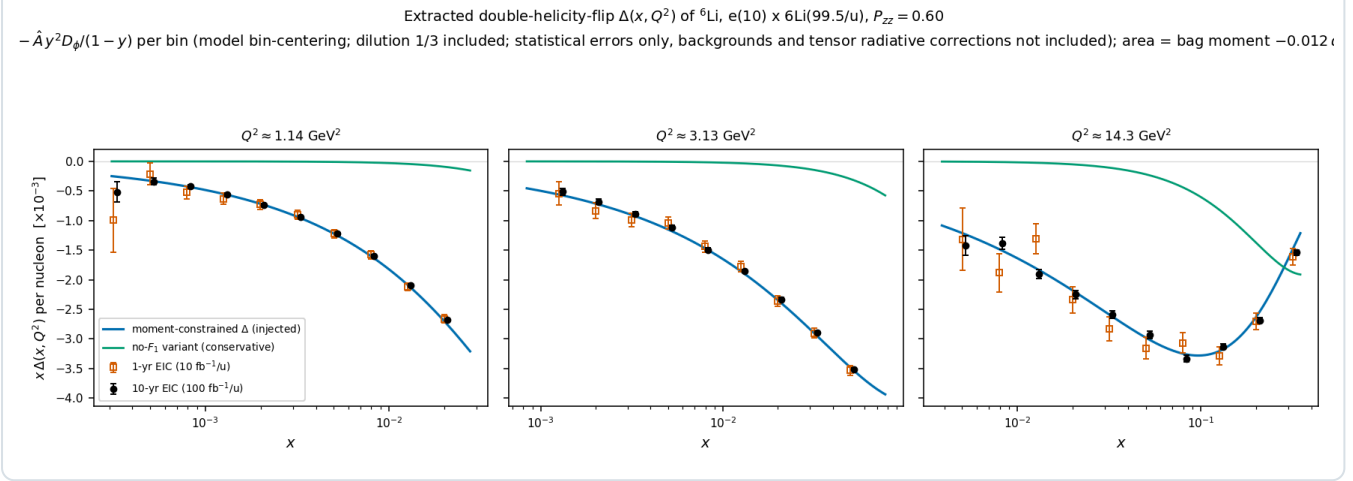
### 5.1 • Truth level

Figure 2 shows the  $\phi'$  pseudo-data in the four super-bins with one-year statistics and the extracted amplitude versus  $x$  along the  $Q^2 = 1.14 \text{ GeV}^2$  slice. The injected amplitudes are  $7.4, 4.4, 9.5$  and  $9.5 \times 10^{-3}$  against one-year errors of  $1.7, 1.4, 2.8$  and  $4.5 \times 10^{-4}$  —  $44, 31, 34$  and  $21\sigma$  — and ten-year errors of  $0.55, 0.45, 0.87$  and  $1.4 \times 10^{-4}$ ; the fitted values agree with the injected ones within 1.5 standard errors. Under the moment constraint the measurement is not a null test: the model-A-versus-B spread is resolved by the  $x$  dependence of the amplitude, and a flat amplitude of  $3 \times 10^{-3}$  per unit  $P_{ZZ}$  would remain a  $3\text{--}11\sigma$  effect per bin even at  $P_{ZZ} = 0.3$ . At ten-year precision the statistical errors sit below the polarimetry scale systematic (3% of an amplitude of  $10^{-2}$  is  $3 \times 10^{-4}$ ), so the ten-year measurement is systematics-limited.



**Figure 2** — inclusive gluonometry at  $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV/u})$ ,  $P_{ZZ} = 0.60$ , model A with dilution 1/3. Left 2x2:  $\phi'$  pseudo-data with one-year statistical errors in the four super-bins ( $Q^2 = 1.1 \rightarrow 14 \text{ GeV}^2$ ,  $x = 0.011\text{--}0.14$ ); injected curve (blue), binned fit (vermillion); annotations give  $\delta A$  for both programmes. Right: extracted amplitude versus  $x$  with one-year (open) and ten-year (filled) errors against the model A, model B and flat- $10^{-3}$  curves.

Figure 3 unfolds the same pseudo-measurements to the structure function,  $x\Delta(x, Q^2)$  per nucleon at the three  $Q^2$  slices of Figure 1, with independent one- and ten-year draws and the model bin-centering factor  $K = \Delta_{\text{model}}(x_c, Q_c^2)/A_{\text{model, bin}}$ ; the area under each curve is the ported bag moment ( $\div 3$  dilution). The best bins reach  $\Delta = -0.135 \pm 0.003$  at  $(x, Q^2) = (0.02, 1.14)$ ,  $-0.070 \pm 0.002$  at  $(0.05, 3.13)$  and  $-0.0047 \pm 0.0004$  at  $(0.32, 14.3)$  in one year, i.e.  $2\text{--}9\%$  relative, and three times better in ten.



**Figure 3** — the extracted  $\Delta(x, Q^2)$ : the amplitudes of Figure 2 unfolded to  $x\Delta$  per nucleon at the three highest-significance  $Q^2$  slices (model bin-centering,  $\delta\Delta = \delta A \cdot |\partial\Delta/\partial A|$ ), with independent one-year (open) and ten-year (filled) pseudo-measurements against the model A and model B curves.

## 5.2 · Reconstructed level

Two of the four super-bins sit at  $y \approx 0.01$ , where the electron method alone gives  $\delta y/y \approx 1.1$ – $1.2$  and the bin is not reconstructible, and the other two at  $y \approx 0.026$ , where it gives  $0.46$ ;  $x$  must come from a hadronic-method  $y$ . The reconstructed-level pseudo-experiments of the companion note [16] pass every event through the lab frame, EMCal energy and track-angle smearing, the head-on transformation, the electron-method  $Q^2$ , a hadronic  $y$ , the mixed-method  $x$ , the analysis cuts on reconstructed quantities, an  $\eta$ -dependent electron identification efficiency and the covariant  $\phi'$ , then form the spin-state ratio of the two fills with a  $3\% \cos 2\phi'$  and  $2\% \cos \phi'$  acceptance harmonic and a  $10^{-3}$  relative-luminosity offset switched on. Table 1 gives the result with the documented 25% Gaussian stand-in for the hadronic  $y$  — the ePIC kinematic-fit study's smearing and the ATHENA value at  $y \approx 0.01$  [17] — and with the  $y$  of the PYTHIA 8 final state through the hadron-side detector response ( $\Sigma$  method, 50 MeV of calorimeter noise), the hadronic scale calibrated from the same simulation as an analysis would, in bins of the reconstructed ( $x, Q^2$ ) [16].

| SUPER-BIN    | 25% HADRONIC-Y STAND-IN |            |      | PYTHIA 8 FINAL STATE, $\Sigma$ METHOD |        |      | $\Delta\hat{A}$ (1 YR) | $\Delta\hat{A}$ (10 YR) |
|--------------|-------------------------|------------|------|---------------------------------------|--------|------|------------------------|-------------------------|
| ( $x, Q^2$ ) | PURITY                  | EFFICIENCY | D    | $\Delta y/y$                          | PURITY | D    | [ $10^{-4}$ ]          | [ $10^{-5}$ ]           |
| 0.028, 1.14  | 0.66                    | 0.42       | 0.92 | 0.32                                  | 0.56   | 1.00 | 1.2                    | 3.7                     |
| 0.011, 1.14  | 0.63                    | 0.59       | 0.99 | 0.22                                  | 0.59   | 1.06 | 0.9                    | 2.9                     |
| 0.071, 3.13  | 0.69                    | 0.38       | 0.90 | 0.29                                  | 0.64   | 0.95 | 1.6                    | 5.2                     |
| 0.141, 14.3  | 0.69                    | 0.66       | 0.96 | 0.15                                  | 0.75   | 0.98 | 3.0                    | 9.3                     |

**Table 1** — the four super-bins at the reconstructed level (mid configuration, spin-state ratio of the  $+0.6 / -1.2$  fills). Purity: fraction of the reconstructed bin's rate that originates in the true bin; efficiency: fraction of the true-bin rate reconstructed in the bin after electron identification; D: rate-weighted,  $\phi'$ -diluted amplitude of the events in the reconstructed bin over that of the true bin;  $\delta y/y$ : 68% half-width of the  $\Sigma$ -method resolution at the spot. The PYTHIA columns have the hadronic scale calibrated in reconstructed ( $x, Q^2$ ) bins; uncalibrated they read purity  $0.42 / 0.53 / 0.49 / 0.68$  and  $D = 0.79 / 0.84 / 0.83 / 0.95$ . The statistical errors are the same for both hadronic- $y$  models to within 7%.

Three conclusions follow. The fit closes on the response-folded truth within the precision of the test: the eight pulls against the reconstructed-bin truth lie within 2.2 standard errors while the acceptance harmonic and the luminosity offset are on, and the  $\sin 2\phi'$  coefficients are consistent with zero. The errors are 0.59–0.69 of the single-state truth-level values, because the  $1.5\times$  gain of the  $m = 0$ -rich fill outweighs the reconstruction efficiency of 0.38–0.66 — presuming the source delivers  $m = 0$ -rich bunches at  $|P_{zz}| = 1.2$  with the same purity as the  $\pm 1$  fill at 0.6. And the real final state costs purity rather than statistics: only 80–92% of the hadronic  $E - p_z$  sum is within the acceptance and above threshold at the four bins, 70–85% through the full response — the rest escapes forward beyond the calorimeters (7–17%, the target-fragmentation side of a  $W \approx 6$ – $10$  GeV system) or lies below threshold — a 13–27% scale bias on  $y_\Sigma$  that an analysis calibrates from its simulation at the reconstructed point. Calibrated, the purities are 0.56–0.75 and the reconstructed-bin amplitude within 6% of the true-



bin value (Table 1); uncalibrated they would be 0.42–0.68 and 5–21% low, and a 1% residual scale error moves  $\hat{A}$  by 0.1–0.6% [16]. The noise floor decides the resolution below  $y \approx 0.02$  — at the mid configuration the  $\Sigma$ -method  $\delta y/y$  at  $y = 0.005$  runs  $0.22 \rightarrow 0.33 \rightarrow 0.54 \rightarrow 1.01$  for  $0 \rightarrow 25 \rightarrow 50 \rightarrow 100$  MeV of calorimeter noise on a hadronic  $E - p_z$  sum of 0.2–0.5 GeV, a regime no published ePIC study covers. Each configuration selects its own four bins: at  $5 \times 40.8$  and  $18 \times 137.5$  they resolve to  $0.38 / 0.23 / 0.24 / 0.11$  and  $0.23 / 0.19 / 0.21 / 0.18$ , so the low- $x$  bins improve with beam energy (the hadronic  $E - p_z = 2E_y$  grows with it) while the  $(0.141, 14.3 \text{ GeV}^2)$  bin degrades,  $0.11 \rightarrow 0.15 \rightarrow 0.18$ , and is best measured at the low configuration. Unfolded to  $\Delta$ , the best reconstructed bins reach statistical errors  $\delta\Delta = 2.5 (2.3) \times 10^{-3}$  at  $Q^2 = 1.14 \text{ GeV}^2$  and  $1.2 (1.1) \times 10^{-3}$  at  $3.13 \text{ GeV}^2$  in one year with the stand-in (PYTHIA)  $y$ , at purities 0.56–0.57 (0.44–0.50).

### 5.3 • Bin centering

Correcting the reconstructed amplitude to a point value of  $\Delta$  with the model's own  $K$  is circular. Because the amplitude is exactly linear in  $\Delta$ , the response can instead be applied as a linear operator and a  $\Delta(x)$  shape fitted through it per  $Q^2$  slice on a bounded grid. Correcting the same pseudo-data with either of two deliberately wrong priors, the model dependence of  $K$  at the four super-bins falls from  $(-4.1, +8.0, -5.5, +4.9)\%$  to  $(-1.5, -1.8, -0.3, +0.5)\%$  with the model B prior and from  $(+8.6, +2.5, +7.3, +3.2)\%$  to  $(+4.6, +3.0, +1.8, -0.9)\%$  with the flat toy, and over the plotted bins of Figure 3 from up to 29% to  $\leq 6\%$ . The  $\Delta$  error bar then carries four terms — statistics, the shape fit, the response Monte Carlo (0.15–1.5% per bin) and the spread over priors the shape family cannot absorb (1–8%) — of which the prior spread dominates in the best-measured bins, where the one-year statistics are  $\lesssim 4\%$ , and statistics elsewhere.

## 6 • The coherent channel

### 6.1 • Detecting the intact recoil

In coherent diffraction the recoil retains the beam's magnetic rigidity —  $A/Z = 2$  exactly, and the diffractive momentum loss  $x_p < 0.01$  changes it by less than 1% — and emerges at  $\theta = p_T/p \lesssim 1$  mrad, inside the beam. Of the far-forward system at IP6 — the B0 spectrometer, the off-momentum detectors, the Roman Pots and the zero-degree calorimeter, with the acceptance windows of Refs. [18–20] — only the pots can see such a recoil, and only where it clears both the  $10\sigma$  beam envelope and the pots' own aperture. Both are constraints on the *angle* at the interaction point, so with an exponential  $t$  distribution of slope  $B$  ( $\approx 50 \text{ GeV}^{-2}$ , §6.2) the tagged fraction is

$$\varepsilon_{\text{tag}} = \exp(-B p_{T,\text{cut}}^2), \quad p_{T,\text{cut}} = \theta_{\text{cut}} A p_u, \quad \theta_{\text{cut}} = \max(10\sigma_\theta, \theta_{\text{aperture}})$$

an exponential in the square of the beam divergence times the beam momentum. Table 2 evaluates it with  $B = 50 \text{ GeV}^{-2}$  at the three  ${}^6\text{Li}$  configurations, using the Yellow Report's divergences per configuration and optics — a  $\gamma$ -matched ion has the proton's  $\beta\gamma$  and hence the proton's divergence at equal normalized emittance, and pays a factor  $\sqrt{2}$  only where it is rigidity-capped, i.e. at the top configuration [13] — and the pots' aperture for an intact  ${}^6\text{Li}$ , measured by tracking one through the ePIC far-forward geometry: the boundary is  $|\theta_x| \gtrsim 2.0 / 1.35 / 1.03$  mrad in the  $5 \times 41 / 10 \times 100 / 18 \times 275$  optics against  $|\theta_y| \gtrsim 2.3\text{--}3.0$  mrad, because the transport images an interaction-point angle onto the pot plane with a horizontal lever  $R_{12} \approx 30.6$  m against a vertical few metres [16]; the scan was made at the proton rigidity of each optics and is applied in angle to the  $\gamma$ -matched beam, which assumes the far-forward magnets scale with the stored rigidity.

| ${}^6\text{Li}$<br>CONFIGURATION | $P_{\text{ION}}$ | $\Sigma_\theta$ H/V, HIGH<br>ACCEPTANCE [MRAD] | $E_{\text{TAG}}, 10\Sigma$<br>ENVELOPE | POT<br>APERTURE  <br>$\theta_x $ | $E_{\text{TAG}},$<br>APERTURE | TAGGING OPTICS:<br>$\Sigma_\theta, B^*$ FACTOR |
|----------------------------------|------------------|------------------------------------------------|----------------------------------------|----------------------------------|-------------------------------|------------------------------------------------|
| $5 \times 40.8$                  | 245<br>GeV       | 220 / 380                                      | $5 \times 10^{-7}$                     | 2.0 mrad                         | $1 \times 10^{-6}$            | 58 $\mu\text{rad}$ , $\times 14.5$             |
| $10 \times 99.5$                 | 597<br>GeV       | 180 / 180                                      | $8 \times 10^{-26}$                    | 1.35 mrad                        | $8 \times 10^{-16}$           | 24 $\mu\text{rad}$ , $\times 58$               |
| $18 \times 137.5$                | 825<br>GeV       | 92 / 92                                        | $4 \times 10^{-13}$                    | 1.03 mrad                        | $2 \times 10^{-17}$           | 17 $\mu\text{rad}$ , $\times 29$               |

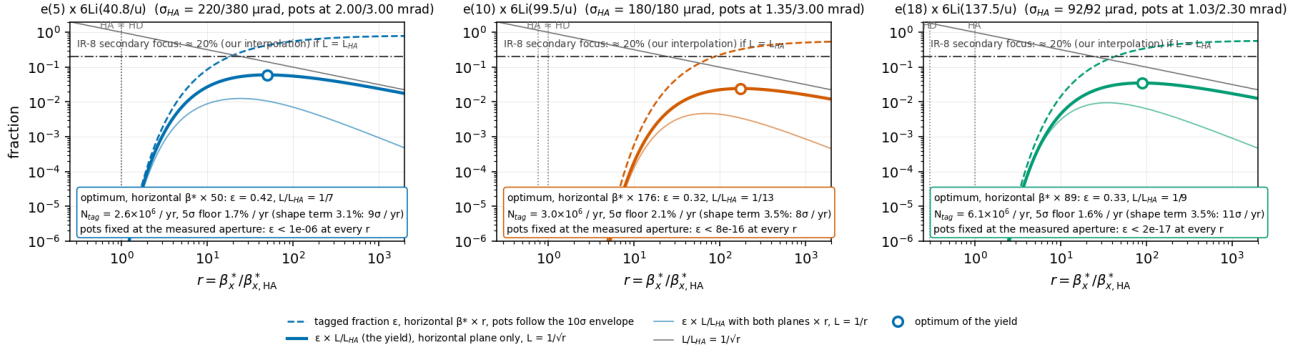
**Table 2** — the coherent tag at IP6 with published inputs. Divergences: Yellow Report Table 10.1 with the  $\gamma$ -matched species step [13]; at 41 GeV the high-acceptance and high-divergence optics are identical, the only configuration where the Yellow Report offers no separate high-acceptance option. Aperture: the rectangular  $|\theta_x| / |\theta_y|$  cutout integrated over the two-dimensional Gaussian, not the circular  $\exp(-B p_T^2)$  of the displayed formula; September-2024 ePIC geometry, one event per scan point, no beam envelope; the pot layout has since changed (16 mm modules, per-energy insertions) and must be re-measured. Tagging optics: the divergence that puts the  $10\sigma$  envelope at  $p_T = 1/\sqrt{B} = 0.14$  GeV, where  $L \times \epsilon_{\text{tag}}$  is maximal and  $\epsilon_{\text{tag}} = 1/e$ , and the  $\beta^*$  increase over the high-acceptance optics it implies; Figure 4 prices the planar-pot version.

The conclusion is unambiguous: with any published optics the coherent intact- ${}^6\text{Li}$  channel does not exist at IP6, at any energy. It exists under two conditions. A *tagging optics*, with  $\beta^*$  raised until the envelope sits at  $p_T = 1/\sqrt{B}$ , where luminosity times acceptance is maximal, gives  $\epsilon_{\text{tag}} = 1/e$  at every configuration; the  $1/e$  column of Table 2 is the circular-isotropic derivation, and priced for planar pots below the optimum is a horizontal-only de-squeeze,  $\beta_x^* \approx 50 \times / 180 \times / 90 \times$  ( $\approx 45$  m at  $5 \times 41$  for a high-acceptance  $\beta_x^*$  of 0.9 m), within the range of the forward-physics optics demonstrated at the LHC — but the pots must then approach horizontally to 0.33, 0.17 and 0.12 mrad where the silicon now sits at  $2.0 / 1.35 / 1.03$ , so the two levers are multiplicative and the second is the near-beam question of Report 4 [21]. Alternatively, the *secondary focus* of the second interaction region lets pots at  $z = 44.0/45.5$  m reach  $p_T \approx 0$ , with published intact-recoil efficiencies of 47% (d), 32% ( ${}^3\text{He}$ ), 29% ( ${}^4\text{He}$ ) and 17.8% ( ${}^7\text{Li}$ ) at top energy, rising steeply at lower energies [22]; no  ${}^6\text{Li}$  case is published and our interpolation is  $\approx 20\%$ .

**Pricing the tagging optics.** The  $\beta^*$  lever is a machine lattice setting, not a far-forward one — it is exactly what separates the Yellow Report's high-acceptance optics from its high-divergence optics — and it is paid for in luminosity:  $\sigma_\theta \propto 1/\sqrt{\beta^*}$  while  $L \propto 1/\beta^*$  per plane, which the Yellow Report's own optics pairs reproduce to 5% (10.0 against  $3.14 \times 10^{33} \text{ cm}^{-2}\text{s}^{-1}$  at 275 GeV for a divergence ratio 119/65) [13]. Because a recoil escapes through the horizontal gap between the pots or through the vertical one, only the plane it escapes through needs the tagging  $\beta^*$ : Figure 4 therefore de-squeezes the horizontal plane alone, holds the vertical at its high-acceptance value and pays  $L/L_{\text{HA}} = 1/r$  rather than  $1/r$ , with the envelope treated as the planar pots see it — a rectangle  $10\sigma_h \times 10\sigma_v$  in angle, the dispersion at the pots ( $D \approx 0.3$  m with the Yellow Report  $\Delta p/p$ ) added to the horizontal side — for pots either following the envelope or fixed at the aperture measured today. With the pots fixed, the acceptance stays below  $10^{-6}$  at any  $\beta^*$ : the far-forward side must move with the optics. With the pots following, the yield peaks at  $r = \beta_x^* / \beta_{x,\text{HA}}^* = 50, 180$  and  $90$  at the three configurations, where  $\epsilon = 0.42, 0.32$  and  $0.33$ , the horizontal envelope sits at 0.33, 0.17 and 0.12 mrad (0.08–0.10 GeV) and the luminosity at  $1/7, 1/13$  and  $1/9.5$  of the high-acceptance value; de-squeezing both planes, the naive reading of Table 2, would cost  $1/24$ – $1/70$  for a fifth of the yield. Taking the placeholder of §3.1,  $10 \text{ fb}^{-1}/\text{u}$  per year, as the high-acceptance luminosity, the tagging optics yields  $2.6 \times 10^6, 3.0 \times 10^6$  and  $6.1 \times 10^6$  tagged events per year —  $2.3 \times, 5.6 \times$  and  $3.9 \times$  below the 0.20 GeV reference tag at that luminosity — and the best super-bin of §6.2 reaches  $5\sigma$  floors of 1.7%, 2.1% and 1.6% per unit  $P_{zz}$  per year. In the optics' own window,  $\langle |t| \rangle = 0.031$ – $0.035 \text{ GeV}^2$ , the shape term of §6.3 is 3.1–3.5% per unit  $P_{zz}$  — a  $9\sigma, 8\sigma$  and  $11\sigma$  measurement per year — and a 1% exotic-glue term reaches  $5\sigma$  in 2.8, 4.4 and 2.6 years. The top configuration is the most productive at equal luminosity, with four times the coherent events of  $5 \times 41$ . If the second interaction region delivered the same luminosity as the high-acceptance optics, its secondary focus at our interpolated  $\approx 20\%$  would still be worth  $3 \times, 8 \times$  and  $6 \times$  the IP6 optimum in yield (Figure 4, dash-dotted line); the factor scales with  $L_{\text{IR8}}/L_{\text{HA}}$ , which is not published, and the published efficiencies rise toward lower energies [22]. Two assumptions carry these numbers and are stated as such: the electron  $\beta^*$  must be raised in step to keep the spot sizes matched, and the far-forward transport of the de-squeezed lattice must remain parallel-to-point at the pots ( $R_{11}\sigma^* \ll R_{12}\sigma_\theta$ , as in TOTEM's high- $\beta^*$  optics) for the envelope to be  $10R_{12}\sigma_\theta$  at all — true of the present lattice at the Yellow Report optics, and the first thing to establish for a de-squeezed one.



A lithium tagging optics at IP6, priced: horizontal  $\beta^*$ -de-squeezed, vertical at high acceptance;  $\sigma_\theta \propto 1/\sqrt{\beta^*}$ ,  $L \propto 1/\beta^*$  per plane, parallel-to-point transport assumed  
 $B = 50 \text{ GeV}^{-2}$ ,  $P_{zz} = 0.60$ ; yields per year at  $10 \text{ fb}^{-1}/u$  in the high-acceptance optics



**Figure 4** — a lithium tagging optics at IP6, priced. Per  ${}^6\text{Li}$  configuration, against  $r = \beta_x^*/\beta_{x,HA}$  with the vertical plane held at high acceptance: the tagged fraction  $\epsilon$  for pots following the  $10\sigma$  envelope (dashed); the luminosity  $L/L_{HA} = 1/\sqrt{r}$  (grey); their product, the tagged yield relative to the high-acceptance luminosity (thick), with the optimum marked and the yield, best-bin  $5\sigma$  floor and shape-term significance per year annotated; and the yield if both planes were de-squeezed by  $r$  (thin). The envelope is a rectangle  $10\sigma_h \times 10\sigma_v$  in angle with the dispersive term added horizontally, which is why the optimum sits far above the circular  $1/e$  point of Table 2;  $B = 50 \text{ GeV}^{-2}$ . With the pots fixed at the aperture measured in the ePIC geometry  $\epsilon$  stays below  $10^{-6}$  at every  $r$  and is stated in each panel rather than drawn. The dash-dotted line is the IR-8 secondary focus at our interpolated  $\approx 20\%$ , drawn as if the second IR delivered the high-acceptance luminosity; HA and HD mark the Yellow Report optics.

**The reference tag of the projections.** Figures 1 and 5 and §6.2 are drawn for a near-beam envelope of  $0.20 \text{ GeV}$  —  $\epsilon_{\text{tag}} = 13.5\%$  (9–20% for  $B \in 40\text{--}60 \text{ GeV}^{-2}$ ), the documented limit of coherent  ${}^3\text{He}$  tagging at IP6 [22] — at the high-acceptance luminosity, which no IP6 optics provides: conservative in the tagged fraction against the 32–42% of the tagging optics and the  $\approx 20\%$  of IR-8, but  $2.3\times$ ,  $5.6\times$  and  $3.9\times$  above what the tagging optics yields at the three configurations. They stand as the physics case for a tag, not as an IP6 projection.

**The configuration we propose.** A lithium tagging optics at  $18 \times 137.5 \text{ GeV}/u$  with the horizontal  $\beta^*$  raised  $\approx 90\times$  over the high-acceptance setting, the vertical plane untouched, and the Roman Pots inserted horizontally to  $\approx 0.12 \text{ mrad}$  ( $\approx 4 \text{ mm}$  at the pot plane if the present lever of  $30.6 \text{ m}$  survived the de-squeeze; the de-squeezed transfer matrix is one of the inputs requested below) — or, at the lowest energy,  $5 \times 40.8 \text{ GeV}/u$  with  $\beta_x^* \approx 50\times$  ( $\approx 45 \text{ m}$  for a high-acceptance  $\beta_x^*$  of  $0.9 \text{ m}$ ) and pots at  $0.33 \text{ mrad}$  — run as a separate running configuration, as the Yellow Report foresees its high-acceptance optics, for a share of the year with which the yields above scale (Table 4, row 8), while the inclusive measurement takes the high-divergence optics. It is a change to the interaction-region lattice, on the electron side as well, and to the pot insertion; whether a horizontal de-squeeze of that size is compatible with the crab cavities and keeps the far-forward transport parallel-to-point is part of what is asked. From the machine it needs the lithium divergence and the high-acceptance  $\beta^*$  per configuration and the far-forward transfer matrix at the pot planes for the de-squeezed lattice; from the detector, the pot approach and the charge identification below; and its cost — a factor 7–13 in luminosity for the duration of the store — is the number to weigh against a second interaction region. Whichever provides the tag, one degeneracy remains: a Roman-Pot hit measures  $A/Z$ , and an intact  ${}^6\text{Li}$ , an  $\alpha$  and a  $d$  at the same velocity share trajectory and timing. The second-detector design answers it with a  $Z^2$ -scaling Cherenkov counter behind the pots [19]; for IP6 no charge-discrimination study exists, though the AC-LGAD readout digitizes an 8-bit charge over four planes and the  $\alpha + d$  pair separates by hit topology [21].

## 6.2 · Rate model

$$\sigma_{\text{coh}} = f_{\text{coh}}(x) \sigma_{\text{DIS}}, \quad f_{\text{coh}}(x) = \frac{f_0}{1 + (x/x_{\text{coh}})^2}, \quad f_0 = 0.04 [0.02\text{--}0.08]$$

No diffractive prediction exists for a light nucleus (the lightest published case is Ca), so the  $f_0$  band brackets HERA ep diffraction (10–15% of DIS [23]) and heavy-nucleus saturation estimates (20–25% [24]); parameterizing  $f_{\text{coh}}$  in Bjorken  $x$  rather than  $x_p$  is optimistic for high-mass diffraction, which the band is meant to cover. The scale  $x_{\text{coh}} = 0.01$  marks where the coherence length  $l_c = 0.105 \text{ fm}/x_p$  falls below the  ${}^6\text{Li}$  size, and the slope  $B = 50 \text{ GeV}^{-2}$  (band 40–60) follows from  $B =$

$\langle r^2 \rangle / 3$  with the matter radius (the charge radius gives  $\approx 57$  [25]; gluonic distributions may be wider [26]), valid below the form-factor minimum at  $|t| \approx 0.31 \text{ GeV}^2$  [27,28]. One year produces  $1.2 \times 10^8$  coherent events, of which the 0.20 GeV tag keeps  $1.7 \times 10^7$  ( $1.7 \times 10^8$  in ten years); the best super-bin,  $x \in [0.0013, 0.0025]$ ,  $Q^2 \in [1, 1.7] \text{ GeV}^2$ , holds  $1.75 \times 10^6$  and gives  $\delta \hat{A} = 1.8 \times 10^{-3}$  (one year) and  $0.6 \times 10^{-3}$  (ten years) on the amplitude per unit  $P_{zz}$ , i.e.  $5\sigma$  floors at 0.9% and 0.3%; at  $P_{zz} = 0.6$  the visible modulations are  $0.6 \times$  these.

### 6.3 • The modulation and its two mechanisms

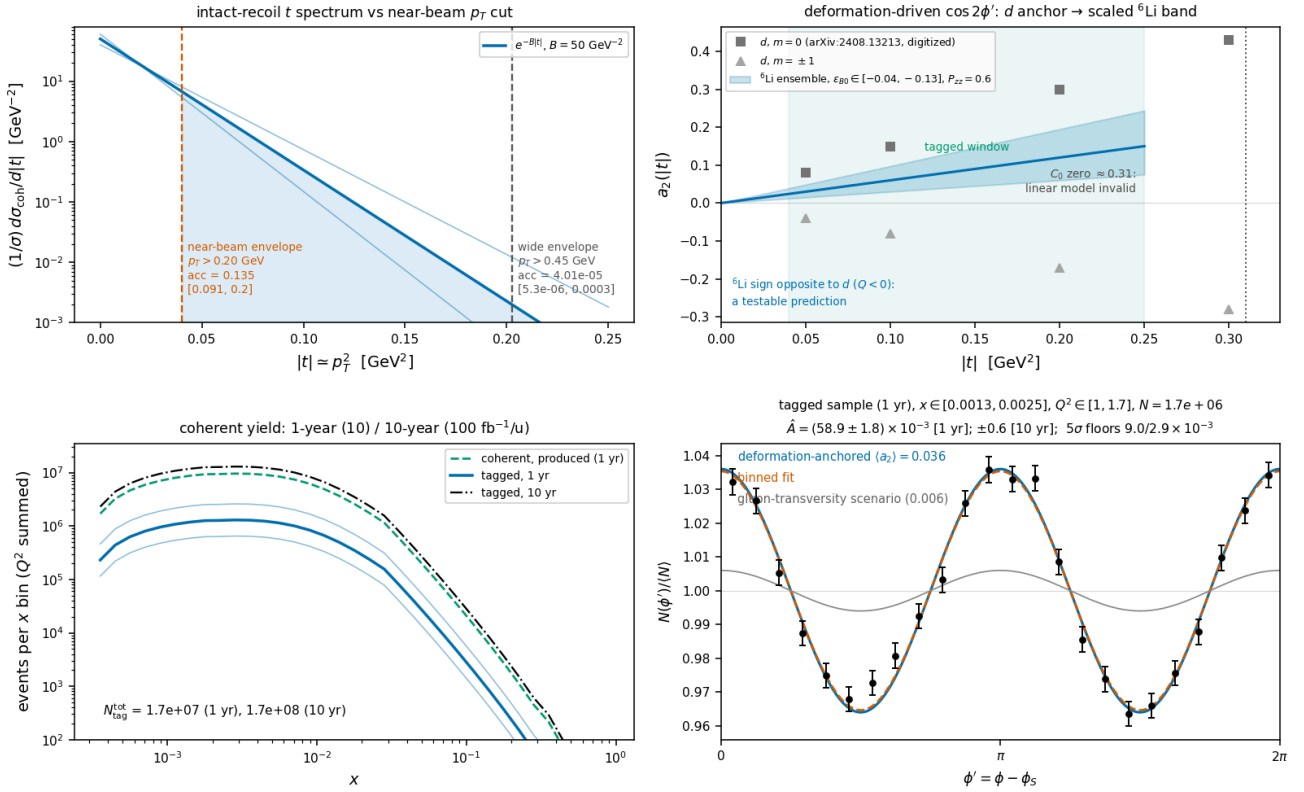
The  $\cos 2\phi'$  modulation of the tagged yield carries two mechanisms. The *deformation* term is scaled from the polarized-deuteron colour-glass calculation of Mäntysaari et al. [10], the only published calculation of polarization-dependent azimuthal coefficients in coherent diffraction on any polarized nucleus (all forward citations checked; the group's subsequent work is unpolarized [29,30]). Their coefficients show  $a_2$  growing linearly in  $|t|$  with  $a_2(m=0) = -2a_2(m=\pm 1)$ , from the 6% D-wave modulating the diffractive slope. For a fill with populations  $p_m$  the ensemble coefficient is proportional to  $P_{zz}$ :

$$a_2(t; P_{zz}) = -\frac{P_{zz}}{4} \varepsilon_{B0} B |t|, \quad \langle a_2 \rangle_{\text{tag}} = a_2(p_{T,\text{cut}}^2 + 1/B) = 0.036 \quad [0.018-0.059]$$

Two conventions matter. The anchor's Eq. (9) expands  $d^2\sigma/d\Phi d|t| \propto 1 + 2\Sigma_n a_n e^{in\Phi}$  with  $\Phi$  the azimuth of the recoil relative to the polarization axis in the  $\gamma^*d$  frame, so the  $\cos 2\Phi$  coefficient is  $2a_2$ : the  $\langle a_2 \rangle$  quoted here — 0.036 at  $P_{zz} = 0.6$ , i.e. 0.060 per unit  $P_{zz}$ , the footing of the  $5\sigma$  floors — and injected in Figure 5(d) understates the deformation modulation by a factor two, and the coherent numbers of this paper are conservative by  $\times 2$ . And the modulation is one of the *recoil* azimuth  $\phi_t - \phi_s$ , not of the electron azimuth used as the axis of Figure 5(d); the reconstructed-level analysis fits both azimuths [16].  $\varepsilon_{B0} \equiv \Delta B_0/B$  is scaled from the deuteron value of 0.21 by two indicators that disagree: the relative quadrupole deformation  $Q/\langle r^2 \rangle$  of  ${}^6\text{Li}$  (0.012) is about one fifth of the deuteron's (0.063) and opposite in sign, while the  $\alpha$ -d asymptotic D/S ratio is disputed — a transfer measurement gives  $\eta = +0.0003(9)$ , consistent with zero, older sub-Coulomb analyses favour  $\eta \approx -0.01$ , and the ab initio no-core shell model with continuum gives  $-0.027$ , the deuteron's own 0.0256 in magnitude [31]. We adopt  $\varepsilon_{B0} \in (-0.04-0.13)$  with default  $-0.08$ , leaving near-zero deformation open; the D/S indicator alone would allow up to the deuteron's  $-0.21$ , and the  $5\sigma$  floors cover detection or exclusion anywhere in that range. The band's width is genuine theoretical uncertainty: even exact few-body calculations miss the  ${}^6\text{Li}$  quadrupole cancellation by a factor  $\approx 3$  [32], and the gluonic deformation need not share the cancellation of the charge sector. The sign flip relative to the deuteron is a testable prediction ( $Q({}^6\text{Li}) < 0$ ), and tensor modulations of comparable size are established at these  $|t|$ : the elastic e-d  $T_{20} = -(0.1-0.3)$  [33] and HERMES  $A_{zz}(d) \approx -1\%$  at  $x \approx 0.01$  [34], the latter attributed to coherent double scattering [35–37]. The flat gluon-transversity term is separately bounded at  $3 \times 10^{-3} - 10^{-2}$  (at  $P_{zz} = 1$ ) from the bare lattice  $\phi$  transversity [6] divided by the  $\sim 10 \times$  NPLQCD deuteron suppression [7], with the maximal Drell–Yan scenario [5] as the ceiling.  $\langle a_2 \rangle$  averages the m-state populations with equal weights; weighting them by their m-dependent rates lowers it by up to 27%, a one-sided shift inside the  $\varepsilon_{B0}$  band.

The two mechanisms separate by shape: deformation is linear in  $t$ , sign-flipped relative to the deuteron and expected to weaken toward small  $x_p$  (JIMWLK evolution washes it out [30]), while exotic glue is flat in  $t$  and persists at small  $x$ , where it connects to the elliptic-gluon programme [38]. In a two-component fit  $a_2(t) = c_{\text{def}} \cdot |t| + a_g$  per  $x_p$  bin, the ten-year floor of 0.3% per unit  $P_{zz}$  lies at or below the flat term's band and an order of magnitude below the deformation term, so both components are expected to be resolved for the 0.20 GeV tag. Incoherent breakup is m-state-blind in this mechanism [10]: it dilutes the modulation but cannot mimic it. And deformation-driven modulations, the polarized-deuteron imaging signal, are bounded on  ${}^6\text{Li}$  by its quadrupole moment,  $Q = -0.0806(6) \text{ fm}^2 \approx Q({}^7\text{Li})/50$  [39]: a sizable flat-in- $t \cos 2\phi$  on  ${}^6\text{Li}$  cannot be a shape effect, and the mechanisms entangled for the deuteron separate in the  ${}^6\text{Li}/d$  comparison — the null test unavailable to the deuteron.

Coherent  $e^6\text{Li} \rightarrow e'X^6\text{Li(g.s.)}$ , transversely tensor-polarized,  $e(10) \times 6\text{Li}(99.5/u)$ ,  $P_{zz} = 0.60$   
 io rates; modulation anchored on the polarized- $d$  CGC calculation, PLB 858:139053; statistical errors only, backgrounds, purity and tensor radiative corrections not in



**Figure 5** — the coherent channel  $e^6\text{Li} \rightarrow e'X^6\text{Li(g.s.)}$  at  $e(10 \text{ GeV}) \times 6\text{Li}(99.5 \text{ GeV}/u)$ ,  $P_{zz} = 0.60$ , for a 0.20 GeV near-beam envelope. (a) Recoil  $t$  spectrum with the slope band and the 0.20 and 0.45 GeV cuts ( $13.5\%$  and  $4 \times 10^{-5}$ ); the panel's optics labels are the proton optics the two cuts were first derived from, neither of which is an available  $^6\text{Li}$  optics ( $\$6.1$ ). (b) Digitized deuteron  $a_2$  per  $m$  state [10] against the scaled  $^6\text{Li}$  ensemble band (sign flip predicted), with the tagged window and the form-factor-zero exclusion. (c) Tagged yield versus  $x$  for both programmes with the  $f_0$  band. (d)  $\phi'$  pseudo-data (one-year statistics) at  $\langle a_2 \rangle = 0.036$  with the flat gluon-transversity scenario (grey); extraction  $\hat{A} \pm 1.8 \times 10^{-3}$  (one year),  $\pm 0.6 \times 10^{-3}$  (ten years) per unit  $P_{zz}$ .

That the analysis closes once a tag exists has been demonstrated with the reconstructed-level chain [16]: the recoil is transported to the pots with divergence smearing and an aperture, the spin-state ratio is formed in the two azimuths and fitted with an acceptance-weighted template basis in which the pot geometry's own  $\langle \cos 2\beta \rangle$  cancels, and reflection-forbidden sin terms serve as null columns. For a 0.73 mrad horizontal approach at the  $5 \times 41$  configuration, the recoil smeared with a 73  $\mu\text{rad}$  divergence (the approach being its  $10\sigma$  envelope, close to the tagging-optics case of Table 2) — acceptance 7.8%,  $1.4 \times 10^6$  tagged events in the lowest of four  $|t|$  bins in one year — the fit recovers the deformation coefficient  $a_t = 0.115 \pm 0.003$ ,  $0.181 \pm 0.004$  and  $0.260 \pm 0.009$  against 0.119, 0.179 and 0.257 injected in the three bins that carry the statistics (the fourth, with  $10^4$  events, returns  $0.41 \pm 0.02$  against 0.354), resolves a pot roll to 6–10 mrad through the null columns, and recovers the flat term to  $\pm 0.0014$  in the lowest bin, where two independent pseudo-experiments return 0.0102 and 0.0151 against 0.010 injected — a residual bias of the template basis at this strongly anisotropic aperture, of the order of the one-year error, that a ten-year measurement would have to remove.

## 6.4 · Backgrounds

| BACKGROUND                                                                                                                                                    | WHY IT FAKES THE TAG                                                                                                                                                                                    | HANDLE                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><math>\alpha + d</math> breakup</b> (threshold 1.474 MeV; the 2.186 MeV $3^+$ state, $\Gamma = 24 \text{ keV}$ , decays $\sim 100\%$ to $\alpha + d$ [40]) | Both fragments at beam rigidity, on the trajectory and velocity of an intact $^6\text{Li}$ : tracking and time-of-flight blind; only $dE/dx$ ( $\propto Z^2$ : 9/4/1) or the two-hit topology separates | Roman-Pot charge discrimination (open item #19, $\$6.1$ ). Fallback: two-component $ t $ fit (coherent $e^{-50 t }$ against a $\sim 10\times$ flatter incoherent slope; crossover at 0.05–0.2 $\text{GeV}^2$ , inside the window [26,41]); $e+\text{Pb}$ benchmark 80–99% incoherent rejection [42] |

|                                                                                             |                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nucleon-emission breakup: $\alpha + p + n$ (3.70 MeV), ${}^3\text{He} + t$ (15.79 MeV) [40] | A fragment could pass unnoticed                                                                      | $p \rightarrow$ off-momentum detectors ( $R = 0.50$ ), $n \rightarrow$ ZDC, ${}^3\text{He} \rightarrow$ Roman-Pot window ( $R = 0.75$ ): vetoable; inefficiency is second order                                                                                                                                                                                |
| ${}^6\text{Li}^*$ 3.5629 MeV ( $0^+$ , $T = 1$ , $\Gamma = 8.2$ eV) [40]                    | Particle-stable ( $\alpha + d$ parity-forbidden); $\gamma$ -decays to a real $A = 6$ , $Z = 3$ track | Isoscalar exchange cannot drive $T = 0 \rightarrow 1$ ; the boosted 0.1–0.4 GeV $\gamma$ lands mostly in the B0 calorimeter (photons $\geq 50$ MeV, 5.5–20 mrad [20]), tail in the ZDC; the 5.366 MeV $2^+ T = 1$ state always emits a nucleon. Every $T = 1$ state is $\gamma$ - or nucleon-vetoable; only $T = 0$ states feed the blind $\alpha + d$ channel |
| Bethe–Heitler / QED Compton                                                                 | Intact nucleus by construction                                                                       | Calculable ( $t$ dependence = charge form factor squared [27,28]); subtract and reuse as an acceptance candle; its charge-quadrupole $\cos 2\phi$ is $< 10^{-3}$ for $ t  < 0.15$ GeV <sup>2</sup> [28,32]                                                                                                                                                     |
| Beam-gas / halo; $e^-$ -side fakes                                                          | Accidental beam-rigidity tracks; $\pi$ / pair-symmetric $e^-$ at high $y$                            | Non-colliding bunches, $\sim 30$ ps timing, vertex association; standard DIS $e^-$ identification plus ( $e^-$ , recoil) kinematic consistency                                                                                                                                                                                                                 |

**Table 3** — backgrounds to the coherent tag, ranked. The  $\alpha + d$  channel is the one that decides the purity.

## 7 • Systematics and assumptions

Two extraction systematics survive the spin-state ratio: the polarimetry scale  $\delta P_{zz}/P_{zz} \approx 3\%$ , common with  $A_{zz}$  and with tensor polarimetry still requiring dedicated development (Lamb-shift or Breit–Rabi-type polarimeters) [8], and radiative corrections for tensor observables, which have not been calculated; the unpolarized QED tail is calculable and doubles as calibration. The detector nuisances have been measured rather than argued, holding the response fixed and varying one thing with common random numbers: a  $\pm 1\%$  electron energy scale moves  $\Delta$  by 0.3–0.9% at the four bins, a hadronic-resolution mismatch (0.30 generated, 0.25 corrected) by 0.5–1.4%, a 1% residual hadronic-scale error by 0.1–0.6%, all above the 0.06–0.28% Monte Carlo floor and below the bin-centering model dependence of §5.3, while an  $\eta$  tilt of the electron-identification efficiency ( $\leq 0.03\%$ ) and the Yellow Report's  $\eta$ -dependent calorimeter table ( $\leq 0.1\%$ ) are unresolved at that floor. In interpretation, a  $\Delta\Delta$ -isobar admixture can mimic  $\Delta$  at leading twist [12] and conventional double scattering generates the low- $x$   $b_1$  [35–37]; the  $Q^2$  lever arm and the  ${}^6\text{Li}/d$  comparison discriminate against both. For the coherent channel, an error in the unpolarized  $\cos 2(\alpha - \beta)$  harmonic  $u_2$  reaches the flat term at first order through  $a_t \delta u_2 \langle \cos 2\beta \rangle$  — a ZEUS- $1\sigma$   $\delta u_2$  [43] moves  $a_e$  by 7–9% at the tagging optics of §6.1 (it was 20% under the slot geometry assumed before the aperture was measured) — and a fill-to-fill change of the pot cutout enters through the acceptance's own  $\langle \cos 2\beta \rangle$ : at the tagging cutout a  $10^{-3}$  change of the horizontal half-width moves  $a_t$  by 0.1–0.8% and a  $10^{-2}$  one by 3–8%, where the earlier slot geometry amplified  $10^{-3}$  to 19% [16]. The pots' envelope must therefore be stable to  $10^{-3}$  between the spin states, or the states must alternate bunch by bunch. Table 4 lists the assumptions and what a wrong one costs.

| # | ASSUMPTION                                                                                                                                                                                                                                                                                                                                      | IMPACT IF WRONG                                                                                         |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1 | A near-beam tag for the intact recoil exists — a tagging optics with pots that follow, or IR-8 (§6.1); the projections take a 0.20 GeV envelope at the high-acceptance luminosity, which no IP6 optics provides; the tagging-optics optimum (a 0.08–0.10 GeV horizontal envelope at 1/7–1/13 of the luminosity) yields 2–6× fewer events (§6.1) | No tag, no coherent channel; the inclusive measurement is unaffected                                    |
| 2 | Coherent fraction $f_0 = 0.04$ [ $\times 2/\div 2$ ], in Bjorken $x$ (§6.2); no diffractive model for a light nucleus exists                                                                                                                                                                                                                    | Yields scale linearly; $\delta A$ as $1/\sqrt{f_0}$                                                     |
| 3 | Bag moment ported to per-nucleon ${}^6\text{Li}$ with dilution 1/3; $x$ shape ours (§4)                                                                                                                                                                                                                                                         | Overall $\Delta$ scale; the literature brackets both directions [7,8]                                   |
| 4 | $\varepsilon_{B0}$ scaled from the deuteron by two indicators disagreeing by $\times 3$ (§6.3)                                                                                                                                                                                                                                                  | Deformation amplitude anywhere in the band, which every plot carries                                    |
| 5 | No event-by-event $Z$ identification in the pots; purity from the $ t $ -shape fit (§6.4)                                                                                                                                                                                                                                                       | $Z$ identification would improve purity; an incoherent slope model that is off grows the fit systematic |

|    |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6  | Linear $a_z(t)$ for $ t  < 0.2 \text{ GeV}^2$ only; the form-factor zero at $0.31 \text{ GeV}^2$ is excluded, not modelled                                                                                                       | The upper tagged window carries unmodelled structure                                                                                                                                                                                                                                                                                                                           |
| 7  | Toy structure functions with grid backends and a placeholder R; no target-mass terms; $g_2 = g_2^{\text{WW}}$ (§3.2)                                                                                                             | Rates to $\times 1.5$ ; shapes are scenarios; the R shift of §3.2 applies to every $\Delta$                                                                                                                                                                                                                                                                                    |
| 8  | All luminosity in transverse-tensor fills at $P_{zz} = 0.6$ , the $m = 0$ -rich fill at $-1.2$ with equal purity, alternating bunch by bunch (§3.3)                                                                              | N scales with the run-plan share and the amplitude with $P_{zz}$ ; without alternation the bias of §3.3                                                                                                                                                                                                                                                                        |
| 9  | Impulse approximation; no final-state interactions; tensor radiative corrections unknown                                                                                                                                         | FSI distorts the recoil spectrum at larger $ t $ ; RC unquantified                                                                                                                                                                                                                                                                                                             |
| 10 | A calorimeter noise floor of 50 MeV on the hadronic $E - p_z$ sum, calorimeters to $ \eta  = 3.7$ (70–85% of it captured at the four bins) and a hadronic scale calibrated from the simulation in reconstructed bins (§5.2) [16] | The purities of Table 1 degrade in proportion to the noise; at 100 MeV the $y \approx 0.01$ bins are lost at the mid configuration. Uncalibrated, the forward escape is a 13–27% scale bias on $y$ and the purities fall to 0.42–0.68; a 1% residual moves $\bar{A}$ by 0.1–0.6%; the ePIC nominal reach of 4.0 recovers a quarter of the escape without changing $\delta y/y$ |

**Table 4** — the assumptions behind the projections, ranked by leverage.

## 8 • Summary and outlook

A transversely tensor-polarized  ${}^6\text{Li}$  beam at the EIC measures  $\Delta(x, Q^2)$  in inclusive DIS at  $21\text{--}44\sigma$  per  $(x, Q^2)$  bin in one year under the moment-constrained scenario, and at  $3\text{--}11\sigma$  per bin for a flat amplitude of  $3 \times 10^{-3}$  per unit  $P_{zz}$  even at  $P_{zz} = 0.3$ , discriminating between interpretations of the bag moment by the  $x$  dependence and extracting  $x\Delta(x, Q^2)$  to 2–9% in the best bins. The result survives reconstruction with the spin-state ratio and a hadronic-method  $y$ , at the price of bin purities that the calorimeter noise floor controls. The measurement makes three demands on the machine that are specific and answerable: tensor polarization delivered as alternating  $m = \pm 1$ -rich and  $m = 0$ -rich bunches, transverse at the interaction point; a  $\cos 2\phi'$  acceptance and a pot envelope stable to  $10^{-4}$  between the spin states, or the bunch-by-bunch alternation that makes any relative-luminosity error a constant; and a tensor polarimetry scale at the few-per-cent level.

The coherent channel carries the same physics with a null test the deuteron cannot offer. It is also a requirement rather than a projection: the intact  ${}^6\text{Li}$  cannot be tagged at IP6 with any published optics, at any energy, and what recovers it is the tagging optics of Figure 4 or the secondary focus of the second interaction region. The tagging optics we propose (§6.1) —  $18 \times 137.5 \text{ GeV/u}$  with the horizontal  $\beta^*$  raised  $\approx 90\times$  and pots at  $0.12 \text{ mrad}$ , or  $5 \times 40.8 \text{ GeV/u}$  at  $\approx 50\times$  and  $0.33 \text{ mrad}$ , the vertical plane untouched, as a separate running configuration — recovers a 33–42% tag at  $1/9.5\text{--}1/7$  of the luminosity: a 9–11 $\sigma$  measurement per year of the shape term with its predicted sign flip against the deuteron, and  $5\sigma$  on a 1% exotic-gluon term in three years; a second interaction region's secondary focus would deliver 3–6 $\times$  more at the same luminosity. The highest-leverage inputs from outside the programme are, on the machine side, a light-ion optics with  $\beta^*$  chosen for tagging and a charge-identification study of the pot readout; on the theory side, a rerun of the IP-Glasma polarized-deuteron setup with an  $\alpha$ -d cluster density, which would replace both  $\epsilon_{B0}$  indicators and the  $f_0$  band at once, a lattice  $\Delta$  moment for a light nucleus, and radiative corrections for tensor observables.

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## Appendix A • Revision history

| DATE       | REVISION                                                                                                                                                                                                                             |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-08-10 | First version (development run 5): money plots 5 and 6 with the moment-constrained $\Delta(x, Q^2)$ and separate one-year / ten-year projections, and the physics note with a verified bibliography; first audit round the same day. |
| 2026-08-11 | Second audit round (formulas against the literature, implementation, references, extraction logic); money plot 7, the extracted $\Delta$ as structure-function data points, added.                                                   |
| 2026-08-17 | Dedicated fact checks of the Sather–Schmidt moment usage and of the far-forward detection chain; phase-space figure with the analysis binning; display mathematics typeset at build time.                                            |
| 2026-08-24 | The anchored coefficient's convention (2a <sub>2</sub> ); the reconstruction-chain report created as the reconstructed-level companion.                                                                                              |



|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-08-25 | The spin-state-sorted ratio as the estimator of the measurement; the near-beam cut stated as angular; the $b_1$ , $b_2$ sign convention; the inversion corrected; the $O(\gamma^2)$ $b_1$ leakage; far-forward parameters updated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2026-08-26 | Development run 8: the PYTHIA 8 hadronic final state; the Roman-Pot aperture measured in the ePIC geometry; the published R1998 fit behind an <code>r_func</code> hook; the forward-folded bin-centering alternative.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2026-08-28 | Numbers of §5.2 and §7 restated after the review of the reconstruction toolchain (Report 2 Appendix A, <code>docs/code_review_2026-08-28.md</code> ): the hadronic-scale calibration is now keyed on the reconstructed $(x, Q^2)$ rather than the event's true cell, and the PYTHIA library was regenerated without PYTHIA's default $\hat{m} \geq 4$ GeV floor, so the calibrated purities of Table 1 read 0.56 / 0.59 / 0.64 / 0.75 ( $D = 1.00 / 1.06 / 0.95 / 0.98$ ) instead of 0.52 / 0.59 / 0.59 / 0.76; the coherent systematics of §7 are quoted at the tagging optics of §6.1 ( $\delta u_z \rightarrow 7\text{--}9\%$ on $a_e$ , a $10^{-3}$ cutout change $\rightarrow \leq 0.8\%$ on $a_t$ ) instead of the superseded slot geometry. No projection of §5.1 or §6 moved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2026-08-27 | Rewritten as a paper on the $\gamma$ -matched beam energies ( ${}^6\text{Li}$ at 40.8 / 99.5 / 137.5 GeV/u, plans/10). All numbers restated from the current scripts: sweet spots at $x = 0.011\text{--}0.14$ , amplitudes $(0.44\text{--}0.95) \times 10^{-2}$ against $\delta A = (1.4\text{--}4.5) \times 10^{-4}$ , the reconstructed-level Table 1, the R-model shifts (+18, +19, +8, $-4\%$ ), the bin-centering scan, $1.2 \times 10^8$ coherent events and $1.7 \times 10^7$ tagged for the 0.20 GeV envelope. The coherent channel restated against the Yellow Report divergences and the measured pot aperture (Table 2): it does not survive at IP6 with any published optics, and the figures are presented for a tagging optics or IR-8 rather than as an IP6 projection; the pre-2026-08-27 6R and measured-aperture results, computed at the superseded rigidity-scaled energies ( $2.7 \times 10^6$ tagged at 50 GeV/u; the low-energy survival at 20.5 GeV/u), are withdrawn and replaced by the analysis closure at a 0.73 mrad approach. Assumptions condensed to Table 4; references numbered in citation order. Later the same day: the tagging optics priced (Figure 4, <code>tagging_optics.py</code> ) — luminosity $\propto 1/\beta^*$ per plane against the tagged fraction, a horizontal-only de-squeeze with the vertical plane untouched, dispersion at the pots and the parallel-to-point transport assumption stated, the pots following or fixed — and the IP6 lithium tagging configuration proposed in §6.1 with its cost. |

Generated from the PolarizedLithiumSim repository: `evgen/scripts/money_cos2phi.py`, `money_cos2phi_coherent.py`, `money_delta_extraction.py`, `phase_space_bins.py`; reconstructed level `money_cos2phi_reco.py` (with `--y-source hfs`, `--unfold-scan`, `--syst-scan`), `hfs_resolution.py`, `coherent_optics_scan.py`, `tagging_optics.py`, `nearbeam_aperture_scan.py`, `nearbeam_reach_gain.py`; models in `polli_fastsim/delta_models.py` and `evgen/polligen/coherent.py`; assembled by `reports/build_report.py`. Every command is in `docs/reproduction_manual.md`. All rates use scenario inputs — bands, not predictions; external numbers are verified against primary sources. Target venue: INT programme on polarized ion beams at the EIC, March 22 – April 2, 2027.